KERTAS CADANGAN PSM
FYP PROPOSAL

SMART LABORATORY BOOKING AND RESOURCE MANAGEMENT SYSTEM FOR FKMP UTHM

NAMA *Name*	:	GANESH RAAJ A/L ARUMUGAM
NO. MATRIK *Matric No.*	:	DI230042
TARIKH *Date*	:	18/10/2025
SUPERVISOR	:	DR. SUHAILA BINTI MOHD. YASIN

Fakulti Sains Komputer dan Teknologi Maklumat
UNIVERSITI TUN HUSSEIN ONN MALAYSIA

ISI KANDUNGAN
TABLE OF CONTENT

1.0 Project Background

Laboratory management refers to the systematic organization and oversight of laboratory resources, including equipment, materials, scheduling, and facility maintenance. It plays a crucial role in supporting teaching, experimentation, and research activities in academic institutions. In universities, especially within faculties such as Mechanical and Manufacturing Engineering, laboratories are essential for hands-on learning, testing, and innovation. These environments require effective coordination of user access, safety protocols, and equipment usage to maintain operational efficiency and uphold educational and research standards. Efficient laboratory management contributes directly to institutional productivity, user satisfaction, and academic excellence (Rihm, 2024; Sun & Gao, 2025).

At present, the Faculty of Mechanical and Manufacturing Engineering (FKMP) at Universiti Tun Hussein Onn Malaysia (UTHM) relies on manual or semi-digital systems for managing laboratory access, bookings, and equipment usage. Laboratory users—including students, researchers, and lecturers—typically request access by filling out paper forms or contacting lab personnel directly via email or in person. Records of equipment usage, asset inventories, and maintenance schedules are maintained using logbooks, spreadsheets, or isolated digital documents. This decentralized approach involves multiple personnel and lacks standardized procedures or shared platforms for consistent data access or coordination (Argento, Rula, & Cowan, 2020; Yin et al., 2022).

The reliance on manual and unintegrated systems has led to several challenges within FKMP laboratories. Scheduling conflicts, double bookings, and underutilization of equipment frequently occur due to the absence of a centralized booking system. Maintenance tracking is often overlooked, increasing the risk of equipment breakdowns and reducing overall operational uptime. Data is scattered across various documents and managed by different individuals, making retrieval slow and prone to errors. Additionally, the lack of automated notifications or digital logs creates inefficiencies in approving bookings and tracking asset conditions. These issues hinder effective decision-making, delay laboratory access, and increase the administrative workload of academic and technical staff (Yin et al., 2022; Rihm, 2024).

To address these challenges, a Smart Laboratory Booking and Resource Management System is proposed for FKMP at UTHM. The system will be a centralized, web-based platform that streamlines laboratory booking, asset tracking, and payment processing through an integrated interface. It will incorporate intelligent automation features such as real-time scheduling, automated reminders, and a virtual assistant powered by natural language processing (NLP) to guide users. The solution will also support proactive maintenance through integrated reporting tools and IoT-based monitoring, allowing staff to track equipment usage and condition more effectively. This approach is expected to improve resource utilization, minimize administrative delays, and promote a more transparent and user-friendly laboratory environment (Sun & Gao, 2025; Rihm, 2024).

2.0 Problem Statement

The current laboratory management process implemented within the Faculty of Mechanical and Manufacturing Engineering (FKMP), Universiti Tun Hussein Onn Malaysia (UTHM), relies heavily on manual and semi-digital procedures for booking, resource allocation, and record-keeping. While functional at a basic level, this system presents several operational challenges that hinder efficiency, transparency, and resource optimization (Yin et al., 2022; Rao et al., 2024).

At present, the laboratory booking process is conducted primarily through manual methods such as physical forms, email requests, or direct communication with laboratory personnel. This fragmented approach often results in scheduling conflicts, overlapping bookings, and delays in approval. Laboratory staff must manually verify availability and record bookings, which increases administrative workload and the likelihood of human error. If this issue remains unresolved, it will continue to limit the efficient utilization of laboratory resources, reduce user satisfaction, and impede the faculty's ability to manage laboratory operations systematically.

The absence of a centralized digital platform for managing laboratory assets and equipment information has led to difficulties in tracking the status, location, and maintenance history of resources. Equipment details are often stored in individual spreadsheets or logbooks maintained separately by each laboratory, resulting in data redundancy and inconsistency. This lack of integration hampers decision-making regarding equipment allocation, maintenance scheduling, and resource planning (Parks et al., 2022; Rao et al., 2024). If not addressed, this problem could lead to asset mismanagement, underutilization of laboratory equipment, and an increased risk of equipment failure due to inadequate maintenance monitoring.

For projects that require material usage fees or external access, payments are typically handled manually, with limited documentation and verification mechanisms. This manual approach complicates financial tracking and makes it difficult to ensure transparency and accountability in transactions. Without a proper digital payment and tracking system, there is a risk of data loss, unauthorized usage, and discrepancies between laboratory usage and recorded income. If this issue persists, it may undermine administrative integrity, reduce financial accountability, and negatively affect the faculty's ability to manage its laboratory operations efficiently.

If these issues are not addressed, FKMP's laboratory operations will continue to face inefficiencies, data fragmentation, and administrative burden. The lack of automation and integration will hinder effective coordination among users and staff, ultimately affecting the quality of teaching, research, and resource management within the faculty. Therefore, the development of a Smart Laboratory Booking and Resource Management System is crucial to ensure centralized data management, improve operational efficiency, and enhance transparency in laboratory utilization and payment processes (Rao et al., 2024; Yin et al., 2022; Yuen et al., 2025).

3.0 Objective

The objectives of this system are to:

1. Analyze and design a Smart Laboratory Booking and Resource Management System using an object-oriented approach.
2. Develop the Smart Laboratory Booking System using web-based approach with XAMPP
3. Test the developed system using alpha and beta testing

4.0 Scope

This project focuses on the development of a Smart Laboratory Booking and Resource Management System within the field of laboratory management and scheduling. The study is conducted at the Faculty of Mechanical and Manufacturing Engineering (FKMP), Universiti Tun Hussein Onn Malaysia (UTHM), where multiple teaching and research laboratories operate under diverse functions that necessitate systematic management of bookings, equipment utilization, and maintenance activities. Effective coordination of these processes is essential to ensure optimal use of laboratory facilities, minimize scheduling conflicts, and maintain the operational efficiency of available resources.

Information required for the system design is obtained through consultations with laboratory technicians, academic staff, and administrative personnel who are directly involved in laboratory coordination, asset maintenance, and resource allocation within FKMP. The users of the proposed system comprise several groups: administrators, responsible for managing user accounts, configuring system parameters, and overseeing overall laboratory operations; laboratory technicians, tasked with equipment registration, maintenance scheduling, and approval of booking requests; academic staff, who reserve laboratory equipment for teaching and research purposes and monitor resource availability; and students or external users, who submit booking requests, process payments, and utilize laboratory facilities according to approved schedules.

The scope of the system is delineated through its core functional modules, as presented in **Table 4.1**. These modules are designed to facilitate key processes such as laboratory booking, asset management, payment handling, maintenance tracking, and reporting. Each user category is granted specific access privileges to the modules relevant to their respective roles and responsibilities, ensuring secure, efficient, and role-based interaction with the system.

Table 4.1: Smart Laboratory Booking System's Module

Module	Description	User
User Authentication and Access Control	Provides secure login and role-based access management for different types of users including administrators, technicians, staff, and students or external users.	Admin, Technician, Staff, Student/ External User
Laboratory and Asset Management	Allows administrators and technicians to register, update, and track laboratory equipment, including specifications, availability, and maintenance history.	Admin, Technician

Table 4.1: Continued

Booking and Scheduling	Enables users to view equipment availability, submit booking requests, and manage time slots. Includes automated approval workflows and conflict prevention.	Staff, Student/ External User, Technician, Admin
Payment and Billing	Manages payment processes for both internal and external users, including automated invoice generation and payment tracking to ensure financial transparency.	Staff, Student/ External User, Admin
Maintenance and Service Management	Allows technicians to log maintenance requests, monitor repair progress, and record service activities to ensure equipment reliability and safety.	Technician, Admin
Reporting and Notification	Generates analytical reports on laboratory usage, payments, and maintenance activities while providing automated notifications for approvals, upcoming bookings, and system alerts.	Admin, Technician, Staff, Student/ External User

5.0 Methodology

This project adopts the Iterative Incremental Model, a software development approach characterized by cycles of planning, design, implementation, testing, and evaluation, which facilitate continuous refinement of the system in response to evolving requirements and stakeholder feedback. Each iteration produces a functional system increment that is progressively enhanced until completion (Nonyelum, 2020; Shaheen et al., 2025). **Figure 5.1** illustrates the Iterative Model applied in this project.

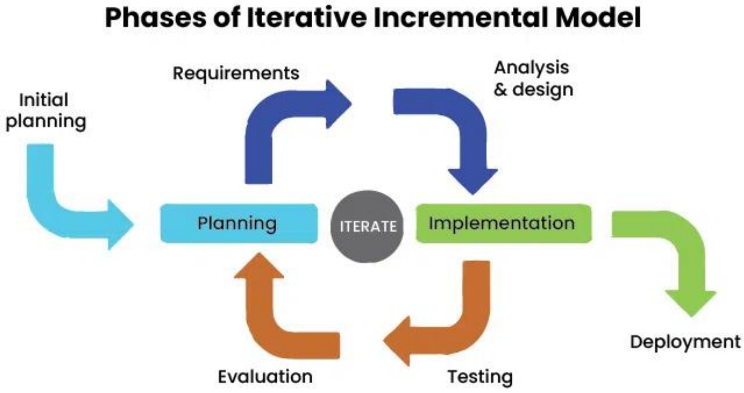

Figure 5.1: Iterative Incremental Model (Nonyelum, 2020; Shaheen et al., 2025)

The process begins with Initial Planning, where project goals are outlined, scope determined, and resources allocated to establish a clear foundation. During Requirements Analysis, detailed needs of faculty administrators, lecturers, and auditors are gathered to specify functionalities such as document management, automated notifications, and dashboard reporting (Shaheen et al., 2025).

The Analysis and Design phase develops the system architecture and user interface, creating data models and workflows integral to supporting teaching and learning audits. Implementation follows, employing Flutter for front-end development and XAMPP with MySQL on the back-end, incrementally integrating features such as secure authentication, file upload, and notification scheduling (Nonyelum, 2020; Shaheen et al., 2025).

Testing is conducted using unit and integration tests to ensure functionality and reliability of each module. Stakeholder feedback during Evaluation guides necessary refinements before final Deployment, where the system is introduced into the faculty environment to support effective management and auditing of teaching documentation (Shaheen et al., 2025; Relia Software, 2024).

By iterating through these phases repeatedly, the model enables adaptive development that handles changing requirements efficiently, ultimately delivering a robust system aligned with user needs (Nonyelum, 2020; Relia Software, 2024).

6.0 Keputusan Jangkaan/ Expected Outcome

The expected outcome of the system focuses on the following purposes:

i. A centralized web-based platform for managing laboratory bookings, equipment, and payments.

ii. Real-time equipment availability and automated notifications for bookings and maintenance.
iii. Accurate tracking of laboratory assets, usage, and maintenance records.
iv. Role-based access control to ensure accountability among all user groups.
v. Reduced administrative workload and human error through process automation.
vi. Improved efficiency, transparency, and reliability in FKMP laboratory management.

7.0 Project Significance

The development of the Smart Laboratory Booking and Resource Management System for FKMP UTHM is significant in enhancing the efficiency and transparency of laboratory operations within the faculty. By integrating booking, payment, and asset management into a centralized platform, the system minimizes dependency on manual processes, which are often time-consuming and prone to error. This digital transformation supports better coordination among administrators, technicians, and academic staff, while providing users with real-time access to equipment availability and maintenance status. The automation of key processes, including notifications and scheduling, contributes to improved time management and optimized utilization of laboratory resources.

Furthermore, the system promotes accountability and data integrity through role-based access control and centralized data storage. It provides a reliable digital record of laboratory activities, which facilitates performance monitoring, reporting, and future planning. For students and external users, the system offers a more convenient and transparent means of booking and payment, while administrators benefit from simplified oversight and decision-making based on accurate information. Overall, this project contributes to FKMP's goal of achieving efficient, data-driven laboratory management practices that support high-quality teaching, learning, and research outcomes.

8.0 Project Planning

The development of the Smart Laboratory Booking and Resource Management System will be carried out over the course of two academic phases, spanning from October 2025 to June 2026. The project will follow an iterative development process, beginning with research, requirement analysis, and prototype design during the early stages, followed by system development, testing, and evaluation in the later stages. To ensure proper tracking of progress, milestones such as proposal preparation, prototype demonstration, system testing, and final presentation are clearly defined within the timeline.

For a detailed breakdown of tasks, durations, and milestones, please refer to the Gantt Chart provided in the Appendix.

9.0 Conclusion

The Smart Laboratory Booking and Resource Management System is developed to improve the existing manual and uncoordinated laboratory management process used at the Faculty of Mechanical and Manufacturing Engineering (FKMP), Universiti Tun Hussein Onn Malaysia (UTHM). The current practice, which relies on paper forms, email communication, and spreadsheets, often results in scheduling conflicts, inefficient equipment utilization, and difficulties in tracking payments and maintenance activities.

This project introduces a centralized, web-based system that integrates laboratory booking, asset management, payment processing, and maintenance tracking into a single platform. Through features such as real-time equipment availability, automated notifications, and role-based access control, the system enhances coordination, minimizes human error, and increases overall operational transparency. Developed using an object-oriented approach and implemented through a web-based environment, the system ensures flexibility, scalability, and continuous improvement through testing and user feedback.

In conclusion, the proposed system provides a more efficient, transparent, and reliable approach to managing FKMP's laboratory resources. It reduces administrative workload, supports informed decision-making, and ensures better accountability in laboratory operations, thereby contributing to FKMP's commitment to improving teaching, learning, and research efficiency through technology-driven management solutions.

Rujukan/ References

Argento, N., Rula, A., & Cowan, D. (2020). Institutional ELN/LIMS deployment: From the trenches. EMBO Reports, 21(10). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7054672/

Nonyelum, O. F. (2020). Iterative and incremental development analysis study of software systems. SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3715318

Parks, D. F., & others. (2022). IoT cloud laboratory: Internet of Things architecture for laboratory device control. *ScienceDirect / Elsevier*. https://www.sciencedirect.com/science/article/pii/S2542660522001007

Rao, A. R., & others. (2024). Designing an Internet of Things laboratory to improve student understanding of secure IoT systems. *Internet of Things and Cyber-Physical Systems*. https://www.sciencedirect.com/science/article/pii/S2667345223000536

Relia Software. (2024, June 30). A complete guide to iterative and incremental development. https://reliasoftware.com/blog/iterative-and-incremental-development

Rihm, S. D. (2024). The digital lab manager: Automating research support. SLAS Technology, 6(1). https://www.sciencedirect.com/science/article/pii/S2472630324000177

Shaheen, A., Alzyadat, W., Alhroob, A., & Asfour, A. (2025). Incremental prioritization using an iterative model for small-scale systems. International Journal of Informatics and Communication Technology, 14(2), 565-574. http://doi.org/10.11591/ijict.v14i2.pp565-574

Sun, H., & Gao, Q. (2025). Design and implementation of an intelligent laboratory management system based on UWB technology. International Journal of Advanced Computer Science and Applications (IJACSA), 16(6). https://doi.org/10.14569/IJACSA.2025.0160678

Yin, J., Zhang, K., Liu, C., & Zhang, M. (2022). Intelligent laboratory management system based on NB-IoT. ACM International Conference Proceedings. https://dl.acm.org/doi/10.1145/3577065.3577087

Yuen, N. H.-Y., Li, C.-H., Kwok, K. C.-M., & Leung, R. (2025). Laboratory Information Management System (LIMS) 4.0: A model to solve the current pain points for commercial electrical and electronic laboratories. *International Journal of Engineering Business Management, 17*. https://doi.org/10.1177/18479790251385743

Appendix

Gantt Chart for Smart Laboratory Booking and Resource Management System

GANTT CHART

Project Name: FKMP TnL Auditing System
Project Start: 6/10/2025
Current Date: 6/10/2025
Weeks in Progress: 0

				6/10/2025	13/10/2025	20/10/2025	27/10/2025	3/11/2025	10/11/2025	17/11/2025	24/11/2025	1/12/2025	8/12/2025
Task	**Start Date**	**End Date**	**Days**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Title Defense & Proposal Approval	6/10/2025	20/10/2025	14										
Literature Review	21/10/2025	30/11/2025	40										
Problem Statement & Objectives	21/10/2025	10/11/2025	20										
Requirement Gathering & Analysis	11/11/2025	5/12/2025	24										
Project Thesis Writing (Chapters 1–3, ongoing)	21/10/2025	20/1/2026	91										
Prototype Development (Initial Build)	15/11/2025	9/2/2026	86										
Prototype Presentation (PSM 1)	10/2/2026	16/2/2026	6										

				15/12/2025	22/12/2025	29/12/2025	5/1/2026	12/1/2026	19/1/2026	26/1/2026	2/2/2026	9/2/2026	16/2/2026
Task	**Start Date**	**End Date**	**Days**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Title Defense & Proposal Approval	6/10/2025	20/10/2025	14										
Literature Review	21/10/2025	30/11/2025	40										
Problem Statement & Objectives	21/10/2025	10/11/2025	20										
Requirement Gathering & Analysis	11/11/2025	5/12/2025	24										
Project Thesis Writing (Chapters 1–3, ongoing)	21/10/2025	20/1/2026	91										
Prototype Development (Initial Build)	15/11/2025	9/2/2026	86										
Prototype Presentation (PSM 1)	10/2/2026	16/2/2026	6										

				23/2/2026	2/3/2026	9/3/2026	16/3/2026	23/3/2026	30/3/2026	6/4/2026	13/4/2026	20/4/2026	27/4/2026
Task	**Start Date**	**End Date**	**Days**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
System Design Refinement	17/2/2026	28/2/2026	11										
Prototype Development (Enhancements &	17/2/2026	15/5/2026	87										
System Testing & Debugging	15/4/2026	31/5/2026	46										
Project Thesis Writing (Chapters 4–5, ongoing)	1/2/2026	25/5/2026	113										
Final Report Compilation	26/5/2026	1/6/2026	6										
Final Submission	2/6/2026	8/6/2026	6										
Final Presentation (PSM 2)	9/6/2026	12/6/2026	3										

				4/5/2026	11/5/2026	18/5/2026	25/5/2026	1/6/2026	8/6/2026	15/6/2026
Task	**Start Date**	**End Date**	**Days**	**31**	**32**	**33**	**34**	**35**	**36**	**37**
System Design Refinement	17/2/2026	28/2/2026	11							
Prototype Development (Enhancements &	17/2/2026	15/5/2026	87							
System Testing & Debugging	15/4/2026	31/5/2026	46							
Project Thesis Writing (Chapters 4–5, ongoing)	1/2/2026	25/5/2026	113							
Final Report Compilation	26/5/2026	1/6/2026	6							
Final Submission	2/6/2026	8/6/2026	6							
Final Presentation (PSM 2)	9/6/2026	12/6/2026	3							